MATH317 · NUMERICAL METHODS

Homework 2 تدريب

Problems 2.1 · Floating-point

محارب

01 الأسئلة

حل كل سؤال في المساحة اللي تحته. الحل الكامل في ملف منفصل.

4(f) SINGLE + DOUBLE

Determine the single-precision and double-precision machine representation of 64.37109375.

SINGLE + DOUBLE **4(c)**

Determine the single-precision and double-precision machine representation of −9876.54321.

HEX TO DECIMAL **8(c)**

Determine the decimal number that has the machine representation $[494F96A0]_{16}$.

SMALL SYSTEM **11**

Enumerate the set of numbers in the floating-point number system that have binary representations of the form $\pm(0.b_1b_2)_2 \times 2^k$, where $k \in \{-1, 0\}$.

UNDERFLOW **15**

In the case of machine underflow, what is the relative error involved in replacing a number x by zero?

ROUNDING **18**

In a five-decimal machine that correctly rounds numbers to the nearest machine number, what real numbers x will have the property fl(1.0 + x) = 1.0?

ROUNDOFF ERROR **21**

(Continuation) What is the relative roundoff error when we round off $2^{-1} + 2^{-26}$ to get the closest machine number?

LOSS OF SIGNIFICANCE 38

Consider a computer that uses five-decimal-digit numbers. Show that if $x = 0.5321487513$ and $y = 0.5321304421$, then the operation $fl(x) - fl(y)$ involves a large relative error. Compute it.